

KNOWLEDGE & RAG · 18 MIN READ

Enterprise RAG Systems & Vector Database Architecture

Production Knowledge Retrieval, Hybrid Search, and Cross-Encoder Re-Ranking

Executive Architecture Summary: Architectural blueprint for building production Retrieval-Augmented Generation (RAG) platforms. Covers semantic chunking strategies, dense vector embeddings, BM25 sparse keyword search, reciprocal rank fusion (RRF), and cross-encoder re-ranking.

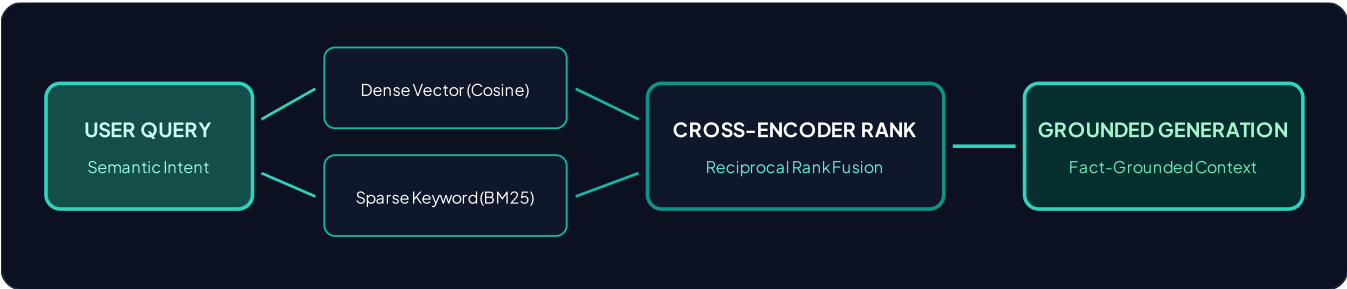
1. Why Naive RAG Fails in Production

Naive RAG—simply splitting text into fixed 500-token chunks, computing cosine similarity, and injecting the top 3 results into the prompt—fails in 40%+ of enterprise use cases. Common failure modes include lost-in-the-middle context degradation, semantic chunk truncation, and keyword mismatch.

- **Semantic Chunking:** Split documents along meaningful paragraph, markdown header, and syntax boundaries.
- **Hybrid Retrieval:** Combine dense vector embeddings with BM25 sparse keyword indices to capture exact terms and semantic concepts.
- **Reciprocal Rank Fusion (RRF):** Merge ranked result lists mathematically to ensure balanced relevance.
- **Cross-Encoder Re-Ranking:** Pass candidate passages through a specialized cross-encoder model to score query-document interaction.

2. The Production Hybrid RAG Pipeline

A multi-stage retrieval architecture filters thousands of documents down to the exact 3–5 most informative passages, maintaining sub-300ms latency.



3. Vector Database Schemas & pgvector Integration

Leverage Neon Serverless PostgreSQL with the pgvector extension for unified relational and vector data storage, eliminating the complexity of running a separate vector database.

```
-- Production pgvector Schema with HNSW Index in Neon Postgres
CREATE EXTENSION IF NOT EXISTS vector;

CREATE TABLE document_knowledge_chunks (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  document_id VARCHAR(255) NOT NULL,
  chunk_index INT NOT NULL,
  content_text TEXT NOT NULL,
  metadata_json JSONB DEFAULT '{}',
  embedding_vector vector(1536), -- OpenAI text-embedding-3-small dimension
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Fast Hierarchical Navigable Small World (HNSW) Index for sub-10ms Cosine Distance
CREATE INDEX ON document_knowledge_chunks
USING hnsw (embedding_vector vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
```

- Metadata Filtering: Pre-filter chunks by tenant ID, date range, or document permissions prior to vector search.
- Context Compression: Strip extraneous markup and summarize retrieved passages to optimize token usage.

4. Evaluation & Continuous Hallucination Mitigation

Measure retrieval quality objectively using RAG Triad metrics: Context Relevance, Groundedness (Faithfulness), and Answer Relevance via synthetic evaluation pipelines.

- Automated Ragas Evaluation: Run continuous regression testing on knowledge base updates.
- Citation Grounding: Force the model to link every factual claim to a specific retrieved chunk ID.